MARWAN EL SAABI GONZÁLEZ

DATA ANALYSIS & MACHINE LEARNING | MSC BIOINFORMATICS CANDIDATE
A Coruña, Spain | marwansaabi@gmail.com | +34 682 448 294 | LinkedIn | GitHub | Portfolio

PROFILE

Biologist and MSc Bioinformatics student (UDC, Feb 2027). I analyse data with Python, SQL and R, and have built personal projects in health data analysis, demand forecasting and machine learning. Looking for a data analyst internship or junior role.

PROJECTS

Clinical Trials Analysis (SQL) | *SQL*, *DuckDB*, *Python*, *Streamlit*, *Plotly*
- SQL analysis of 600,000+ clinical trials (AACT, 40+ tables): ETL from PostgreSQL to DuckDB, reusable views and KPIs.
- Cleaned inconsistent records and classified 39,000 free-text stop reasons; 38% of failures were due to recruitment.

Retail Demand Forecasting | *Python*, *LightGBM*, *Pandas*, *Streamlit*
- Forecast weekly SKU demand 4 weeks ahead from 1M+ transactions using LightGBM with lag, price and calendar features.
- Beat naive, seasonal-naive and moving-average baselines by 4.8% wMAPE under rolling-origin backtesting.

Alzheimer's Drug Classifier (ML Virtual Screening) | *Python*, *Scikit-learn*, *XGBoost*, *RDKit*
- Curated 9.7k ChEMBL IC50 records into 4.3k unique molecules, encoded as 1024-bit Morgan fingerprints.
- Benchmarked RF, SVM, XGBoost and KNN with scaffold-split CV: 0.92 AUROC (a random split inflates it to 0.96).

Melanoma Detection (Classical Computer Vision) | *Python*, *OpenCV*, *Scikit-learn*, *Scikit-image*
- Automatic segmentation of dermoscopic lesions via Otsu, K-means and watershed, selected by IoU against ground truth.
- Extracted 40 shape, color and texture (LBP/GLCM) features; SVM reached 90.2% accuracy and 84% melanoma sensitivity.

PROFESSIONAL EXPERIENCE

- **Freelance AI Data Quality Analyst / TELUS Digital AI Data Solutions (remote)** | Sept 2026 to Present
- **Delivery Driver (Part-time) / Teigal Logística (Amazon Delivery Partner)** | June 2026 to Present
- **Production Operator / Stellantis España S.L, Vigo** | Oct 2024 to Oct 2025
- **Seasonal Farmworker (Grape Harvest) / Iman Temporing ETT, S.L, for Lagar de Fornelos** | Sept 2024

EDUCATION

University of A Coruña (Universidade da Coruña), A Coruña | Sept 2025 to Feb 2027
MSc in Bioinformatics Applied to Health Sciences
Master's thesis (in progress): information system on US packaged-food composition (UML, relational database, SQL, Power BI).
University of Vigo (Universidade de Vigo), Vigo | Jul 2024
BSc in Biology (Specialization: Biotechnology)

AREAS OF EXPERTISE

- **Programming & Data:** Python (NumPy, Pandas, OpenCV), R (Tidyverse, ggplot2, DESeq2), SQL, Excel, MATLAB, Bash.
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Neural Networks, Cross-Validation, Forecasting, LLM APIs (Gemini).
- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, Conda, Nextflow, High-Performance Computing (SLURM).

LANGUAGES

Spanish and Galician (native), English (B2, CertAcles)

PUBLICATION

Alarcón-Ríos L., **Saabi González M.**, Velo-Antón G. (2026). Melanism in salamanders: effects of insularity, sex and size on dorsal colouration in *Salamandra salamandra*. *Journal of Zoology*. doi:10.1111/jzo.70133